Youssef Yasser Elhakie

AI & Software Engineer

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GitHub: github.com/itz-youssef **Portfolio:** youssefyasser.me **Location:** Cairo, Egypt 12513 (Open to Relocation & Remote)

Profile

AI & Software Engineer with approximately six months of internship and training experience, building computer vision and generative AI applications using Python, PyTorch, and FastAPI. Work includes real-time object detection and tracking, deep learning model training, and RAG applications. Combines Machine learning and backend skills to build and integrate AI-powered services.

Education

Cairo University Cairo, Egypt
Bachelor's in Computer Science and Artificial Intelligence *Sep. 2023 – Expected July 2027*

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Signals & Systems, Supervised & Unsupervised Learning, Reinforcement Learning, Robotics & Autonomous Multiagent Systems.

Professional Experience

Aitronix On-site, New Cairo, Egypt
Artificial Intelligence Intern | *Certificate* *Jun. 2026 – Aug. 2026*

- Worked on computer vision and LLM tasks, including model fine-tuning and reducing hallucinations, and assisted with **n8n** automation workflows for WhatsApp bots.
- Deployed **YOLOv8s** models onto enterprise IP cameras via RTSP, achieving **25+ FPS** spatial monitoring across **4** concurrent streams.

Computer Vision Intern *Apr. 2026 – Jun. 2026*

- Built and optimized real-world CV pipelines, integrating **Vision Science** and **RAG** for enhanced context retrieval.
- Engineered **Behavioral AI Theft Detection** system and used **VLMs (Moondream)** for worker safety monitoring Project.

Projects

Pie: AI-Powered Marketing Kit Generator | pie-ai-eg.vercel.app | *VLMs, Groq LPU, RAG, FastAPI* August 2026

- **Generative AI & RAG:** Built a full-stack AI platform using **Vision-Language Models (VLMs)** to transform product screenshots into structured marketing kits in under **60 seconds**.
- **High-Speed Inference:** Integrated **Groq-hosted LLM inference** with a custom **RAG** pipeline to retrieve contextual information for marketing content generation.

Botfolio: AI Portfolio Chatbot Platform | nexus-eight-ivory.vercel.app | *Python, FastAPI, FAISS, Groq, SQLite* June 2026

- Built a multi-tenant AI chatbot platform Experimented by 10 users, with JWT authentication, password hashing, resume uploads, customizable bots, and shareable public chat links.
- Implemented RAG with per-chatbot FAISS indexes and Groq-powered responses, SQLite data storage, Stripe Checkout and billing webhooks, and an HTML/CSS/JavaScript dashboard for chatbot configuration and user/plan management.

YAI : Video Safety Analyzer | github.com/itz-youssef/OmniTracker | *PyTorch, FastAPI, YOLOv10x, CLIP* May 2026

- Built a five-stage video analysis pipeline combining object detection, open-vocabulary classification, multi-object tracking, behavioral anomaly scoring across six threat categories, and monocular depth estimation.
- Integrated YOLOv10x, CLIP, ByteTrack, AnomalyTransformer, and Depth Anything v2 through an asynchronous FastAPI backend with job queues and WebSocket result streaming to a React frontend hosted on Netlify.

Smart Agriculture Crop Disease Detection | **Kaggle** | *ViT, Attention-CNN, DCGAN, PyTorch* Feb - May 2026

- **Automated Diagnosis & Kaggle Bronze Medal:** Built a robust disease classification system trained on **54,000+** images across **15** categories. Benchmarked **14** architectures, with a custom **CNN** achieving **97.5%** validation accuracy. Awarded a **Kaggle Bronze Medal** for the comprehensive model evaluation notebook.

Behavioral AI Theft Detection System | github.com/itz-youssef/Theft-Detection-Model | *YOLOv11x, SAHI, LSTM* April 2026

- **Robust Detection & Tracking:** Engineered a real-time pipeline actively monitoring **30** high-value classes using **YOLOv11x** and **SAHI** (**1000**×**1000** tiling), integrated with **ByteTrack** and **ResNet18** ReID.
- **Behavioral Classification:** Extracted **17** dynamic spatial features (grab motion, joint angles) to train an attention-based PyTorch **LSTM** network, classifying suspicious temporal movements across occlusions.

Technical Skills

Programming: Python, Java, C++, SQL, JavaScript, C#; OOP, Algorithms, Design Patterns, Git/GitHub

Machine & Deep Learning: PyTorch, TensorFlow/Keras, Scikit-learn, Hugging Face, Pandas, NumPy

Computer Vision: OpenCV, YOLO, Tracking, Pose Estimation, Instance Segmentation, SAHI, Anomaly Detection, Fine tuning

Data & Automation: Data Cleaning, Preprocessing, Visualization, Feature Engineering, Augmentation, Model Evaluation, Data Engineering, n8n

Generative AI: LLMs, NLP, VLMs, RAG, AI Agents, CLIP, LangChain, LlamaIndex, ChromaDB, Pinecone

Full-Stack Development: FastAPI, REST APIs, WebSockets, HTML/CSS, React, PostgreSQL, Supabase

Cloud & MLOps: Docker, Kubernetes, CI/CD, GitHub Actions, MLflow, GCP, AWS (SageMaker, Bedrock, S3, EC2), Azure

AI-Assisted Development: OpenAI Codex, Claude Code, OpenCode, GitHub Copilot, Cursor; debugging, refactoring

Professional Mentorship & Training Programs

Deloitte NextStep Mentorship Program Member Oct. 2026 – Nov. 2026

Deloitte Innovation Hub *Remote, Cairo, Egypt*

- Participated in technical webinars covering **AI, machine learning, and data engineering**, exploring industry use cases and AI practices through interactive discussions and mentor Q&A.

AI Trainee Huawei ICT Academy Program | *Certificate* Sep. 2026 – Oct. 2026

Huawei ICT Academy-Egypt & Ministry of Youth and Sports of Egypt *Remote, Cairo, Egypt*

- Completed instructor-led training covering **machine learning, deep learning, and PyTorch**, including **CNNs, RNNs, and Transformers**.
- Completed mentor-guided practical AI exercises, reinforcing AI development concepts through hands-on implementation and problem-solving.
- Passed the program's final **screen-switch-monitored HCIA Exam** with **866 points**, answering **52 of 60 questions correctly**.

Computer Vision Engineer Trainee | *Certificate* Jul. 2026 – Aug. 2026

National Telecommunication Institute (NTI) *Hybrid, Cairo, Egypt*

- Collaborated on assembling a dataset of **10,000+ images** from multiple datasets available on **Roboflow** for driver fatigue and attention monitoring.
- Co-Developed a **Driver Fatigue & Attention Monitoring System (GitHub)**, using the collected dataset to train and compare **3 model architectures**, achieving **90% validation accuracy** and **95% training accuracy**, with real-time drowsiness detection at **30+ FPS** using **MediaPipe** and **MobileNetV2**.
- Built and evaluated a custom **TensorFlow/Keras CNN** for five-grade knee osteoarthritis classification using **8,260 X-ray images** across training, validation, and test sets, incorporating data augmentation and class-weighted training.

Competitions

Contestant | *Certificate* July 2024

Egyptian Collegiate Programming Contest (ECPC) *Alexandria, Egypt*

- Collaborated in a team (Ctrl z) to solve algorithmic problems in **C++** under strict contest time constraints, applying data structures, debugging solutions, and prioritizing problems.

Freelance Experience

Outlier June 2026

Freelance AI Response Evaluator *Remote*

- After strict screening for eligibilty checking I Evaluated and rated conversational AI responses for **reinforcement learning from human feedback (RLHF)** across naturalness, engagement, utility, and audio quality, providing detailed, evidence-based justifications to support model improvement.

Professional Certificates & Courses

Huawei Certified ICT Associate (HCIA-AI V4.0) | *Huawei ICT Academy (ID: ICT20261004014238)* Oct. 2026

AWS: Becoming a Machine Learning Engineer | *Manara (ID: 1792443688-D3705093E310C95B)* Oct. 2026

Microsoft Applied Skills: Secure AI Solutions, Cloud Security & Entra ID | *Microsoft* Sep. 2026

Deep Learning Specialization | *DeepLearning.AI (Andrew Ng) – Coursera* Feb. 2026

Languages

Arabic: Native **English**: Professional Working Proficiency (B2+ / Duolingo English Test: 130)